Derivation and Implementation Notes

Gallimore & Liao — Quantum Computing for Heavy Quarkonium Spectroscopy
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Companion document for the `quarksim` codebase

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1 The Physics Problem

1.1 Charmonium and the Cornell Potential

The system under study is a charm quark (c) and anti-charm quark ($\bar{c}$) bound together by the strong force. At the energy scales of charmonium ($\sim$ 1–3 GeV), a non-relativistic potential model provides a good description.

The interaction is the **Cornell potential** (Eq. 1 of the paper):

$$V(r) = -\frac{\kappa}{r} + \sigma r \tag{1}$$

where:

- $-\kappa/r$: color Coulomb attraction from one-gluon exchange (short-range),
- σr: linear confinement from the QCD string tension (long-range).

1.2 Parameters

The numerical values used in the paper (Section II.A):

Symbol	Value	Description
κ	0.4063	Dimensionless color Coulomb coupling
$\sqrt{\sigma}$	441.6 MeV	String tension
μ	637.5 MeV	Reduced mass ($\mu = m_c/2$, $m_c \approx 1275$ MeV)
ω	562.9 MeV	QHO oscillator frequency

Natural units ($\hbar = c = 1$) are used throughout.

1.3 The Hamiltonian

In the center-of-mass frame, the Hamiltonian for relative motion is (Eq. 2):

$$H = T + V = -\frac{\nabla^2}{2\mu} + V(r) \tag{2}$$

The goal is to find the eigenvalues (energy levels) and eigenstates of H on a quantum computer.

Code: `hamiltonian.py`, constants in `CHARMONIUM_PARAMS`.

2 Basis Expansion: QHO States

2.1 Choice of Basis

The Hamiltonian is expanded in **spherical quantum harmonic oscillator (QHO) s-wave states** $\{|n\rangle\}_{n=0}^{N-1}$, with wavefunctions (Eq. 4):

$$\langle r|n|r|n\rangle = (-1)^n \sqrt{\frac{2\,n!}{b^3\,\Gamma(n+3/2)}} \exp\left(-\frac{r^2}{2b^2}\right) L_n^{1/2}\left(\frac{r^2}{b^2}\right) \tag{3}$$

where:

- $b \equiv (\mu\omega)^{-1/2}$ is the oscillator length,
- $L_n^{1/2}$ are generalized Laguerre polynomials.

Why QHO states? They form a complete, orthogonal basis. The Gaussian envelope naturally matches bound-state wavefunctions. In the final result, the ground state is $\sim$ 98.6% in the $|0\rangle$ orbital, confirming an excellent choice of basis.

2.2 Truncation to $N = 3$

The Hilbert space is truncated to $N = 3$ states: $|0\rangle$, $|1\rangle$, $|2\rangle$. This yields a 3×3 Hamiltonian matrix encodable in 3 qubits.

By the **Hylleraas–Undheim–MacDonald theorem** (Appendix A of the paper), the eigenvalues of H_N are upper bounds on the true eigenvalues of the full (infinite-dimensional) H.

3 Matrix Elements

3.1 Kinetic Energy (Eq. 5)

The kinetic energy operator is **tridiagonal** in the QHO basis—it only couples adjacent states n and $n \pm 1$:

$$\langle m|T|n\rangle = \frac{\omega}{2}\left[\left(2n + \frac{3}{2}\right)\delta_{mn} - \sqrt{n\left(n + \frac{1}{2}\right)}\,\delta_{m,\,n-1} - \sqrt{(n+1)\left(n + \frac{3}{2}\right)}\,\delta_{m,\,n+1}\right] \tag{4}$$

For $N = 3$, the kinetic energy matrix is:

$$T = \frac{\omega}{2}\begin{pmatrix} 3/2 & -\sqrt{3/2} & 0 \\ -\sqrt{3/2} & 7/2 & -\sqrt{5} \\ 0 & -\sqrt{5} & 11/2 \end{pmatrix} \tag{5}$$

Code: `hamiltonian.kinetic_element(m, n, omega)`

3.2 Potential Energy (Eqs. 6–7)

The matrix elements of r and r^{-1} are evaluated using hypergeometric functions.

Coulomb term $\langle m|r^{-1}|n\rangle$ (Eq. 7):

$$\langle m|r^{-1}|n\rangle = (-1)^{m+n}\frac{4\,b^{-1}}{\pi(1+2n)}\sqrt{\frac{\Gamma(m+3/2)\,\Gamma(n+3/2)}{m!\,n!}}\;{}_3F_2\big(\tfrac{1}{2}, 1, -m;\ \tfrac{3}{2}, \tfrac{1}{2} - n;\ 1\big) \tag{6}$$

Linear confinement $\langle m|r|n\rangle$ (Eq. 6):

$$\langle m|r|n\rangle = (-1)^{m+n}\frac{4b}{\pi(1-4n^2)}\sqrt{\frac{\Gamma(m+3/2)\,\Gamma(n+3/2)}{m!\,n!}}\;{}_2F_1\big(2, -m;\ \tfrac{3}{2} - n;\ 1\big) \tag{7}$$

The full potential matrix element is:

$$V_{mn} \equiv \langle m|V|n\rangle = -\kappa\,\langle m|r^{-1}|n\rangle + \sigma\,\langle m|r|n\rangle \tag{8}$$

Code: `hamiltonian.potential_element(m, n, b, kappa, sigma)`. Uses `mpmath.hyp3f2` and `mpmath.hyp2f1` for numerical evaluation.

4 Jordan-Wigner Transformation

4.1 Second Quantization (Eq. 3)

The Hamiltonian in second-quantized form:

$$H_N = \sum_{m,n=0}^{N-1} h_{mn}\,a_m^\dagger a_n \tag{9}$$

where $h_{mn} = \langle m|T + V|n\rangle$ and $a_n^\dagger$, a_n create/destroy a $c\bar{c}$ pair in orbital $|n\rangle$.

The physical states (single-particle sector):

$$|001\rangle = \text{orbital 0 occupied (QHO ground state)}$$
$$|010\rangle = \text{orbital 1 occupied}$$
$$|100\rangle = \text{orbital 2 occupied}$$

4.2 Jordan-Wigner Mapping (Eqs. 8–9)

The creation and annihilation operators map to qubit (Pauli) operators:

$$a_n^\dagger = \frac{1}{2}\left(\prod_{j<n} Z_j\right)(X_n - iY_n) \tag{10}$$

$$a_n = \frac{1}{2}\left(\prod_{j<n} Z_j\right)(X_n + iY_n) \tag{11}$$

From these, one derives:

$$a_n^\dagger a_n = \frac{1}{2}(I - Z_n) \qquad \text{(diagonal)} \tag{12}$$

$$a_m^\dagger a_n + a_n^\dagger a_m = \frac{1}{2}(X_m X_n + Y_m Y_n) \qquad \text{(adjacent, } |m-n| = 1) \tag{13}$$

$$a_0^\dagger a_2 + a_2^\dagger a_0 = \frac{1}{2}(X_0 Z_1 X_2 + Y_0 Z_1 Y_2) \qquad \text{(non-adjacent, } Z \text{ string)} \tag{14}$$

4.3 The 10-Term Pauli Hamiltonian (Eqs. 11–20)

Substituting the Jordan-Wigner expressions into (9):

$$H_3^0 = \frac{1}{2}(h_{00} + h_{11} + h_{22})\, I \otimes I \otimes I \tag{15}$$

$$H_3^1 = -\frac{h_{00}}{2}\, I \otimes I \otimes Z \tag{16}$$

$$H_3^2 = -\frac{h_{11}}{2}\, I \otimes Z \otimes I \tag{17}$$

$$H_3^3 = -\frac{h_{22}}{2}\, Z \otimes I \otimes I \tag{18}$$

$$H_3^4 = \frac{h_{01}}{2}\, I \otimes X \otimes X \tag{19}$$

$$H_3^5 = \frac{h_{12}}{2}\, X \otimes X \otimes I \tag{20}$$

$$H_3^6 = \frac{h_{01}}{2}\, I \otimes Y \otimes Y \tag{21}$$

$$H_3^7 = \frac{h_{12}}{2}\, Y \otimes Y \otimes I \tag{22}$$

$$H_3^8 = \frac{h_{02}}{2}\, X \otimes Z \otimes X \tag{23}$$

$$H_3^9 = \frac{h_{02}}{2}\, Y \otimes Z \otimes Y \tag{24}$$

where $h_{mn} = T_{mn} + V_{mn}$.

Key structural properties:

- The XX and YY terms from the same orbital pair always share equal coefficients.
- The XZX and YZY terms also share the same coefficient $h_{02}/2$.
- The identity term is a constant energy offset.

Qiskit convention: In a Pauli string `'ABC'`, `A` acts on qubit 2, `B` on qubit 1, `C` on qubit 0. So $Z_0 =$ `'IIZ'`, $X_0X_1 =$ `'IXX'`, $X_0Z_1X_2 =$ `'XZX'`.

Code: `hamiltonian.build_pauli_hamiltonian()`. The function computes h_{mn} from scratch using `kinetic_element()` and `potential_element()`, then constructs all 10 terms directly. The function `validate_pauli_hamiltonian()` verifies that the 8×8 Pauli matrix matches the 3×3 physical matrix in the 1-particle subspace.

5 The UCC Ansatz

5.1 Unitary Coupled Cluster Operator (Eq. 22)

The variational ansatz rotates the reference state $|001\rangle$ (one particle in the ground orbital) into a general single-particle superposition:

$$U(\theta, \varphi) = \exp\left\{\theta(a_1^\dagger a_0 - a_0^\dagger a_1) + \varphi(a_2^\dagger a_0 - a_0^\dagger a_2)\right\} \tag{25}$$

5.2 Reparametrization (Eq. 23)

Defining $\alpha \equiv \sqrt{\theta^2 + \varphi^2}$ and $\sin\beta \equiv \theta/\alpha$:

$$\boxed{|\psi(\alpha, \beta)\rangle = \cos\alpha\ |001\rangle + \sin\alpha \sin\beta\ |010\rangle + \sin\alpha\cos\beta\ |100\rangle} \tag{26}$$

This is the most general state with exactly one particle (up to global phase), parametrized by just 2 real numbers (α, β).

5.3 Circuit Decomposition (Fig. 1)

The paper provides a low-depth gate decomposition using RY rotations, CNOT gates, and one X gate. The circuit has depth 5 and uses 8 gates total.

Starting from $|000\rangle$, the sequence of gates is:

1. $RY(\beta)$ on qubit 1
2. $RY(2\alpha)$ on qubit 0
3. CNOT: qubit 0 $\to$ qubit 2
4. CNOT: qubit 2 $\to$ qubit 1
5. X on qubit 0
6. $RY(-\beta)$ on qubit 1
7. CNOT: qubit 2 $\to$ qubit 1
8. CNOT: qubit 1 $\to$ qubit 2

The correctness can be verified by tracing through the 8-component statevector and confirming it produces (26).

Code: `ansatz.build_ansatz()` returns a `QuantumCircuit` with parameters `alpha` and `beta`. The function `ansatz_state(alpha, beta)` computes the statevector analytically for validation.

6 The VQE Algorithm

6.1 Variational Principle (Eq. 21)

For any trial state $|\psi(\alpha, \beta)\rangle$:

$$\langle\psi(\alpha, \beta)|H_3|\psi(\alpha, \beta)\rangle \geq E_0 \tag{27}$$

where E_0 is the ground state energy. VQE finds the parameters (α^*, β^*) that minimize the left-hand side.

6.2 Measurement Strategy

On a quantum computer, one cannot directly measure $\langle H_3 \rangle$. Instead:

1. Prepare $|\psi(\alpha, \beta)\rangle$ using the ansatz circuit.
2. Measure each of the 9 traceless Pauli terms (Eqs. 16–24) separately.
3. Combine: $\langle H_3 \rangle = \sum_{i=0}^{9} c_i \langle P_i \rangle$ where P_i is the Pauli operator and c_i the coefficient.

Each Pauli string expectation value requires measuring in the appropriate basis (Hadamard for X, $S^\dagger$–Hadamard for Y, computational basis for Z).

6.3 Classical Optimization

The paper uses the **COBYLA** algorithm (Constrained Optimization By Linear Approximation), a derivative-free method well-suited for the noisy cost function landscape of quantum hardware. It typically converges in ∼50–100 iterations for this system.

Code: `quarksim.simulation.run_vqe()` implements the full VQE loop using exact statevector simulation (Qiskit `Statevector`). Uses `scipy.optimize.minimize` with COBYLA.

7 Error Mitigation: Zero-Noise Extrapolation

The paper employs **zero-noise extrapolation** (ZNE, Section II.E) to correct for decoherence on real hardware.

7.1 The Paper's Model

Under a global depolarizing channel with layer success rates r_i, the noisy expectation value of a traceless operator scales as:

$$\langle H_3^i \rangle(\lambda) = \left\langle \tilde{0} \middle| \tilde{U}^\dagger H_3^i \tilde{U} \middle| \tilde{0} \right\rangle \; r^\lambda, \qquad 1 \le i \le 9 \tag{28}$$

where λ is a noise scaling parameter ($\lambda = 1$ is the base circuit). The key idea: by running the circuit at multiple noise levels, one can fit the exponential decay and extrapolate back to $\lambda = 0$ (the noiseless limit).

7.2 Circuit Folding

To increase λ without changing the logical operation, the circuit is **folded**: U is replaced by $U \cdot (U^\dagger U)^n$, which is logically identical to U but has $(2n+1)\times$ the depth. The additional gate layers accumulate proportionally more noise, giving an effective scale factor $\lambda = 2n + 1$.

Implementation subtlety: When simulating ZNE, the circuit transpiler must be run with `optimization_level=0`. Otherwise, the transpiler recognizes $U^\dagger U = I$ and simplifies the folded circuit back to U, defeating the purpose of the noise amplification.

7.3 Our Improved Fit Model

The paper fits each traceless term with $y(\lambda) = A \cdot r^\lambda$. In our simulated experiments, this model fails for terms with small noiseless values (specifically $H_3^5 = X_1 X_2$ and $H_3^7 = Y_1 Y_2$), exactly as the paper reports:

> "the plots for $\langle H_3^5 \rangle$ and $\langle H_3^7 \rangle$ conform especially poorly to the global depolarizing model."

The root cause: depolarizing noise on the measurement basis-rotation gates (H for X, $S^\dagger H$ for Y) creates a **constant positive bias** that is independent of λ. For terms with large noiseless values ($\langle P \rangle \sim 1$), this bias is negligible. But for terms with small noiseless values ($\langle P \rangle \sim 0.03$), the bias dominates and the measured values actually *increase* with λ — the opposite of what (28) predicts.

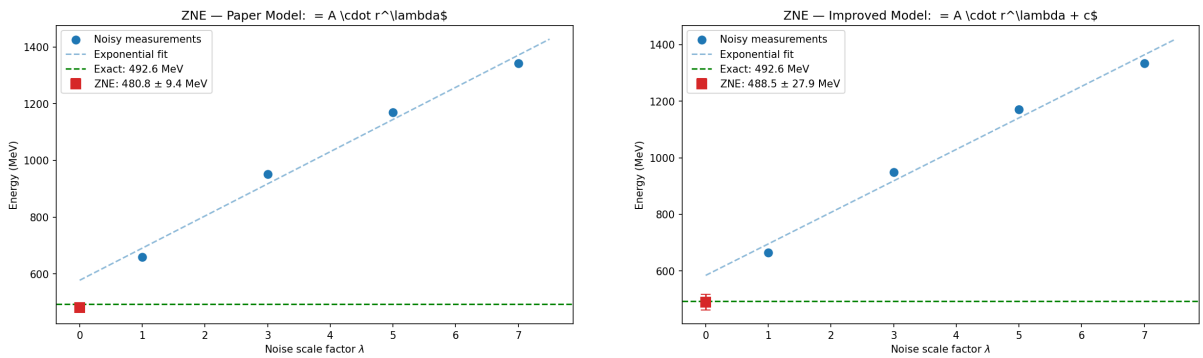
We address this with an improved fit model that includes a constant offset:

$$\boxed{y(\lambda) = A \cdot r^\lambda + c} \tag{29}$$

where c captures the measurement noise floor. At $\lambda = 0$, the noiseless estimate is $y(0) = A + c$. This model correctly handles both well-behaved terms (where $c \approx 0$) and bias-dominated terms (where c absorbs the offset).

7.4 Comparison: Paper's Model vs. Improved Model

Figures 1 and 2 compare the two ZNE models side by side. The data is from a simulated run with 2% per-gate depolarizing noise and 65,536 shots per Pauli term, at noise scale factors $\lambda = 1, 3, 5, 7$ via global circuit folding.



(a) Paper's model: $y = A \cdot r^\lambda$

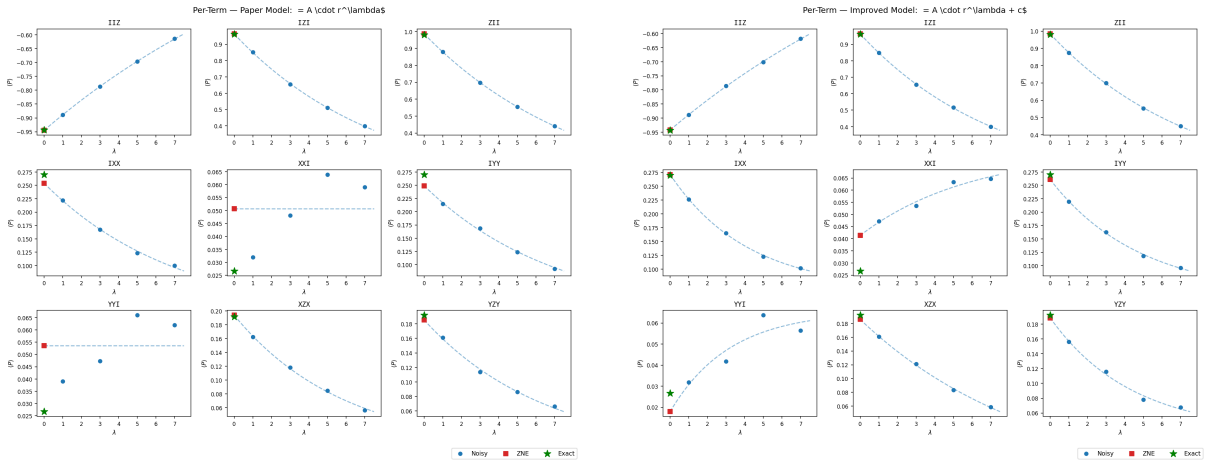
(b) Improved model: $y = A \cdot r^\lambda + c$

Figure 1: Total energy vs. noise scale factor λ under the two ZNE models. Both extrapolate to $\lambda = 0$ (red square) to estimate the noiseless energy. Green dashed line = exact diagonalization (492.6 MeV). The improved model incorporates the measurement noise offset c.

Method	Energy (MeV)	Error vs. exact
Noisy VQE ($\lambda = 1$, 2% per-gate)	$\sim$665	+172 MeV
ZNE — paper's model ($A \cdot r^\lambda$)	481 ± 9	−12 MeV
ZNE — improved model ($A \cdot r^\lambda + c$)	489 ± 28	−4 MeV
Exact diagonalization	492.6	—
Paper's ZNE (real hardware)	502 ± 98	+9 MeV

The paper's model undershoots the exact energy because the flat fits on `XXI` / `YYI` overestimate those terms at $\lambda = 0$. The improved model is closer to exact but has wider error bars (3 fit parameters vs. 2 gives less-constrained extrapolation). Both are substantial improvements over the uncorrected noisy VQE.

Code: `quarksim.simulation.run_zne()` implements the full pipeline. Use `--noise 0.02 --zne` when running `quarksim.gallimore.run`.



(a) Paper's model: $y = A \cdot r^\lambda$

(b) Improved model: $y = A \cdot r^\lambda + c$

Figure 2: Per-Pauli-term ZNE (analogous to the paper's Fig. 3). Blue dots = noisy measurements at each λ; red squares = ZNE extrapolation to $\lambda = 0$; green stars = noiseless exact values. Key difference visible in the `XXI` and `YYI` panels: the paper's model produces a flat fit (the data *increases* with λ, violating $r \leq 1$), while the improved model correctly separates the decaying signal from the constant measurement bias.

8 Excited States via Orthogonalization

The paper generalizes VQE to excited states (Section II.D) by orthogonalizing with respect to the ground state.

Given the ground state $|\psi(\alpha_0, \beta_0)\rangle$, the first excited state is found by minimizing $\langle H_3 \rangle$ over all states $|\psi(\alpha, \beta)\rangle$ satisfying:

$$\langle \psi(\alpha_0, \beta_0) | \psi(\alpha, \beta) | \psi(\alpha_0, \beta_0) | \psi(\alpha, \beta) \rangle = 0 \tag{30}$$

The orthogonality constraint reduces the 2-parameter space to a single parameter γ (Eqs. 37–39):

$$\cos \alpha_1 = -\sin \alpha_0 \cos \gamma \tag{31}$$

$$\sin \alpha_1 \sin \beta_1 = \cos \alpha_0 \sin \beta_0 \cos \gamma + \cos \beta_0 \sin \gamma \tag{32}$$

$$\sin \alpha_1 \cos \beta_1 = \cos \alpha_0 \cos \beta_0 \cos \gamma - \sin \beta_0 \sin \gamma \tag{33}$$

Appendix B of the paper proves that errors in the excited state estimates are bounded by the ground state error: $|\delta_1| \leq \delta_0$ and $|\delta_2| \leq \delta_0$.

For the 2nd excited state (3S), once two orthonormal states are known the third is fully determined as the cross product in the 3D amplitude space.

Code: `ansatz.orthogonal_state(alpha0, beta0, gamma)` constructs the orthogonal state parametrized by γ. `simulation.run_vqe_excited()` optimizes over γ via bounded scalar minimization. `ansatz.third_state()` computes the 2nd excited state from the first two via cross product.

9 Results

9.1 Paper's Expected Values

State	Exact diag. (MeV)	Noiseless VQE (MeV)	Real hardware (MeV)
1S (ground)	492.6	493 ± 1	502 ± 98
2S (excited)	1210.8	1212 ± 2	—

Ground state wavefunction (Eq. 36):

$$|\psi_{1S}\rangle = -0.9858\,|001\rangle - 0.1369\,|010\rangle - 0.09722\,|100\rangle \tag{34}$$

with $\alpha_0 = 3.31$, $\beta_0 = 0.95$.

9.2 Our Code's Results

Quantity	Value
E_0 (exact diag.)	492.63 MeV
E_1 (exact diag.)	1210.84 MeV
E_2 (exact diag.)	2151.64 MeV
E_0 (VQE, statevector)	492.63 MeV
α (VQE)	3.311
β (VQE)	0.955
Wavefunction	$-0.986\,\|0\rangle - 0.137\,\|1\rangle - 0.097\,\|2\rangle$

All results match the paper to within numerical precision.

10 Bugs Fixed from Original Code

The original implementation (in `QC of Heavy Quarkonium/QHO_elemetnts.py`) contained two critical bugs:

10.1 Bug 1: Missing ω in off-diagonal kinetic terms

The off-diagonal kinetic energy contributions to the Pauli coefficients were missing the factor of ω:

```
# WRONG (original code):
Tk["4"] = -np.sqrt(3/2)            # missing * omega
Tk["5"] = -np.sqrt(5)              # missing * omega
```

The diagonal terms (indices 0–3) correctly included `omega`, but indices 4–7 did not. Since $\omega = 562.9$ MeV, this made the off-diagonal kinetic contributions $\sim$563$\times$ too small.

10.2 Bug 2: Wrong matrix element in H_3^9

```
# WRONG (original code):
H["9"]["coeff"] = (1/2) * (Tk["9"] + Vnm["00"])  # uses V_00
# CORRECT:
H["9"]["coeff"] = (1/2) * (Tk["9"] + Vnm["02"])  # should be V_02
```

The YZY term (Eq. 24) uses the same matrix element h_{02} as the XZX term (Eq. 23). The original code erroneously used V_{00} instead of V_{02}.

10.3 Fix Approach

Rather than patching the hardcoded dictionary, the new code computes the full matrix h_{mn} from `kinetic_element()` and `potential_element()`, then derives all 10 Pauli coefficients directly from the matrix. This eliminates the possibility of such transcription errors and is validated by `validate_pauli_hamiltonian()`.

11 Code–Equation Mapping

Paper equation	Code location	Description
Eq. (1): $V(r)$	`hamiltonian.potential_element()`	Cornell potential
Eq. (5): $\langle m\|T\|n\rangle$	`hamiltonian.kinetic_element()`	Kinetic energy
Eqs. (6–7)	`hamiltonian.potential_element()`	Potential integrals
Eqs. (8–9)	`hamiltonian.build_pauli_hamiltonian()`	Jordan-Wigner
Eqs. (11–20)	`hamiltonian.build_pauli_hamiltonian()`	10-term decomposition
Eq. (22): $U(\theta, \varphi)$	`ansatz.build_ansatz()`	UCC circuit
Eq. (23): $\|\psi(\alpha, \beta)\rangle$	`ansatz.ansatz_state()`	Analytical state
VQE loop	`quarksim.simulation.run_vqe()`	Optimization

12 Running the Simulation

```
# Install
uv venv && source .venv/bin/activate
uv pip install -e .

# Run
python -m quarksim.gallimore.run --seed 42

# With options
python -m quarksim.gallimore.run --maxiter 500 --output results/
    gallimore
```

Output files:

- `vqe_result.json` — full simulation record
- `convergence.png` — VQE energy vs. iteration
- `energy_levels.png` — energy level comparison
- `wavefunction.png` — ground state wavefunction amplitudes